University of Cape Town

Department of Mathematics and Applied Mathematics

Mathematics II **Advanced Calculus (2AC)**

Class Test 2: **9th May 2023**

Time: 1.5 hrs **Marks available: 36** **Full Mark: 34**

Solutions:

Section A **Full Mark: 26**

In this section full answers are expected. Marks will be deducted for incomplete solutions.

1. Find the points of maximum and minimum of the function $f(x, y, z) := -x + y^2 - z^2$ over the sphere $x^2 + y^2 + z^2 = 1$.

We use the method of the Lagrange multiplier by finding x, y, z, λ such that

$$\begin{cases} -1 = 2\lambda x, \\ 2y = 2\lambda y, \\ -2z = 2\lambda z, \\ x^2 + y^2 + z^2 = 1, \end{cases} \checkmark \quad \Leftrightarrow \quad \begin{cases} -1 = 2\lambda x, \\ 2y(\lambda - 1) = 0, \\ 2z(\lambda + 1) = 0, \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

From the second and third equations, we find that:
For $\lambda \neq \pm 1$, we get $y = z = 0$ and hence, $x = \pm 1$ from the fourth equation and $\lambda = \pm\frac{1}{2}$. So we obtain, the points

$$\boxed{(-1, 0, 0)\ 1/2 \checkmark \quad \text{and} \quad (1, 0, 0)\ 1/2 \checkmark}.$$

For $\lambda = 1$, we get $x = -\frac{1}{2}$ from the first equation, $z = 0$ from the third equation and $y = \pm\frac{\sqrt{3}}{2}$ from the fourth equation. Hence, obtain the points

$$\boxed{\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right)\ 1/2 \checkmark \quad \text{and} \quad \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)\ 1/2 \checkmark}.$$

For $\lambda = -1$, we get $x = \frac{1}{2}$ from the first equation, $y = 0$ from the second equation and $z = \pm\frac{\sqrt{3}}{2}$ from the fourth equation. Hence, obtain the points

$$\boxed{\left(\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right)\ 1/2 \checkmark \quad \text{and} \quad \left(\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right)\ 1/2 \checkmark}.$$

Now, we evaluate the function f at the points above and find that

$$f(-1, 0, 0,) = 1,\ f(1, 0, 0) = -1,\ f\left(-\frac{1}{2}, \pm\frac{\sqrt{3}}{2}, 0\right) = \frac{1}{2} + \frac{3}{4} = \frac{5}{4}, \quad f\left(\frac{1}{2}, 0, \pm\frac{\sqrt{3}}{2}\right) = -\frac{1}{2} - \frac{3}{4} = -\frac{5}{4}.$$

So, $\left(\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right)$ and $\left(\frac{1}{2}, 0, \frac{\sqrt{3}}{2}\right)$ are the points of minimum 1/2 ✓ with minimum value equal to $-\frac{5}{4}$ while $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right)$ and $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0\right)$ are the points of maximum 1/2 ✓ with maximum value equal to $\frac{5}{4}$.

[5]

2. Find the cartesian equation of the tangent plane to the level surface

$$\mathcal{S} := \{(x,y,z) \in \mathbb{R}^3\colon\ \sin(x+y)+\sin(x+z)+\sin(y+z) = 1+\sqrt{2}\}$$

at the point $(\frac{\pi}{4}, \frac{\pi}{4}, 0)$.

Set $f(x,y,z) := \sin(x+y)+\sin(x+z)+\sin(y+z)$. The tangent plane to the level surface $\mathcal{S}$ at the point $(\frac{\pi}{4}, \frac{\pi}{4}, 0)$ has normal vector $\nabla f(\frac{\pi}{4}, \frac{\pi}{4}, 0)$. 1/2 ✓ **This mark could also be awarded later if the student has shown an understanding that the vector gradient is the normal vector to the tangent plane!**
Now,

$$\begin{cases} f_x(x,y,z) = \cos(x+y)+\cos(x+z), \ \text{1/2 ✓} \quad f_y(x,y,z) = \cos(x+y)+\cos(y+z), \ \text{1/2 ✓} \\ f_z(x,y,z) = \cos(x+z)+\cos(y+z)\,. \ \text{1/2 ✓} \end{cases}$$

So,

$$\begin{aligned} \nabla f(\tfrac{\pi}{4}, \tfrac{\pi}{4}, 0) &= \Big(f_x(\tfrac{\pi}{4}, \tfrac{\pi}{4}, 0),\ f_y(\tfrac{\pi}{4}, \tfrac{\pi}{4}, 0),\ f_z(\tfrac{\pi}{4}, \tfrac{\pi}{4}, 0)\Big) \\ &= \Big(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \sqrt{2}\Big)\,. \checkmark \end{aligned}$$

Hence, the tangent plane has cartesian equation

$$\begin{aligned} \Big(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \sqrt{2}\Big)\cdot\Big(x-\frac{\pi}{4},\ y-\frac{\pi}{4},\ z-0\Big) = 0 \ &\Leftrightarrow\ \frac{1}{\sqrt{2}}\Big(x-\frac{\pi}{4}\Big)+\frac{1}{\sqrt{2}}\Big(y-\frac{\pi}{4}\Big)+\sqrt{2}(z-0) = 0 \\ &\Leftrightarrow\ \boxed{x+y+2z = \frac{\pi}{2}}\,. \checkmark \end{aligned}$$

[4]

3. Let $\mathcal{S}$ be the surface described by the vector function $\vec{r}\colon D = [0,\pi]\times[-\frac{\pi}{4}, \frac{\pi}{4}] \to \mathbb{R}^3$ defined by

$$\vec{r}(s,t) := (\cos(2s),\ -\sin s + t,\ 2s - \cos t) \text{ for every } (s,t)\in D\,.$$

(a) Show that $\vec{r}_s(s,t)\times\vec{r}_t(s,t) = (-\cos s\sin t - 2,\ 2\sin(2s)\sin t,\ -2\sin(2s))$ for every $(s,t)\in D$.

$$r_s(s,t) = (-2\sin(2s),\ -\cos s,\ 2) \ \text{1/2 ✓} \ \text{and} \ \vec{r}_t(s,t) = (0,\ 1,\ \sin t)\,. \ \text{1/2 ✓}$$

So,

$$\vec{r}_s(s,t)\times\vec{r}_t(s,t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -2\sin(2s) & -\cos s & 2 \\ 0 & 1 & \sin t \end{vmatrix} = (-\cos s\sin t - 2,\ 2\sin(2s)\sin t,\ -2\sin(2s))\,. \checkmark$$

[2]

(b) Show that the surface $\mathcal{S}$ described by the vector function $\vec{r}$ is regular or smooth.

It follows from (a) that $\vec{r}_s$ and $\vec{r}_t$ are continuous. ✓
On the other hand, if there are (s,t) in the interior of D such that $\vec{r}_s(s,t)\times\vec{r}_t(s,t) = (0,0,0)$, then we will get

$$\begin{cases} -\cos s\sin t - 2 = 0, \\ 2\sin(2s)\sin t = 0, \\ -2\sin(2s) = 0 \ \Rightarrow\ s = \frac{\pi}{2} \in (0,\pi), \ \text{1/2 ✓} \ \Rightarrow\ \cos s = 0\,. \end{cases}$$

Hence,

$$\vec{r}_s(\pi/2, t) \times \vec{r}_t(\pi/2, t) = (-2, 0, 0) \neq (0, 0, 0), \ \ 1/2 \checkmark .$$

One could also use only the first equation as follows:

$$|\cos s \sin t| \leq 1 \ \Rightarrow \ -\cos s \sin t \neq 2 \ \Rightarrow \ -\cos s \sin t - 2 \neq 0 \checkmark$$

[2]

(c) Write down the cartesian equation of the tangent plane to the surface $\mathcal{S}$ at the point $\left(0, -\frac{\sqrt{2}}{2}, \frac{\pi - 2}{2}\right)$.

We first find the values of s and t such that

$$\vec{r}(s,t) = \left(0, -\frac{\sqrt{2}}{2}, \frac{\pi-2}{2}\right) \quad \Leftrightarrow \quad \begin{cases} \cos(2s) = 0 \ \Rightarrow \ s = \frac{\pi}{4}, \frac{3\pi}{4}, \\ -\sin s + t = -\frac{\sqrt{2}}{2}, \\ 2s - \cos t = \frac{\pi - 2}{2} . \end{cases}$$

For $s = \pi/4$, we get from the second equation that $t = 0$ and the third equation is also satisfied.
For $s = 3\pi/4$, we get from the second equation that $t = 0$ but the third equation is not satisfied.
Therefore, we find that $s = \pi/4$ 1/2 ✓ and $t = 0$. 1/2 ✓
Award the marks if students just guessed!

Thus, the normal vector of the tangent plane to the surface $\mathcal{S}$ at the point $\left(0, -\frac{\sqrt{2}}{2}, \frac{\pi-2}{2}\right)$ is

$$\vec{r}_s(\pi/4, 0) \times \vec{r}_t(\pi/4, \, 0) = (-2, \, 0, \, -2) \, . \ \ 1/2 \checkmark$$

Hence, the tangent plane has cartesian equation

$$\begin{aligned} & (x - 0, y + \sqrt{2}/2, z - (\pi - 2)/2) \cdot (-2, 0, -2) = 0 \\ \Leftrightarrow \quad & -2 \times (x - 0) + 0 \times \left(y + \frac{\sqrt{2}}{2}\right) - 2 \times \left(z - \frac{\pi - 2}{2}\right) = 0; \ \ 1/2 \checkmark \\ \Leftrightarrow \quad & x + z = \frac{\pi - 2}{2} . \end{aligned}$$

[2]

4. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) := y \sin(3x) + y^2 \qquad \forall (x, y) \in \mathbb{R}^2 .$$

(a) Find **all** the critical points of the function f.

We seek points (x, y) such that $f_x(x, y) = 0$ and $f_y(x, y) = 0$. We find that

$$\begin{cases} 3y \cos(3x) = 0, \ \ 1/2 \checkmark \\ \sin(3x) + 2y = 0 . \ \ 1/2 \checkmark \end{cases}$$

We find from the second equation that $y = 0$ or $\cos(3x) = 0$.

If $y = 0$, then we get from the second equation that

$$\sin(3x) = 0 \quad \Leftrightarrow \quad 3x = k\pi,\ k \in \mathbb{Z} \quad \Leftrightarrow \quad x = \frac{k\pi}{3},\ k \in \mathbb{Z}\,.$$

Thus, we get the points

$$\boxed{\left(\frac{k\pi}{3}, 0\right) \quad k \in \mathbb{Z}} ✓$$

If $\cos(3x) = 0$ then we get $3x = \frac{\pi}{2} + n\pi$ with $n \in \mathbb{Z}$, that is $x = \frac{\pi}{6} + \frac{n\pi}{3}$ with $n \in \mathbb{Z}$. Hence, $\sin(3x) = (-1)^n$. So, we get from the second equation that

$$(-1)^n + 2y = 0 \quad \Leftrightarrow \quad y = -\frac{(-1)^n}{2}\,.$$

We obtain the points (according to whether n is odd or even)

$$\boxed{\left(\frac{\pi}{6} + \frac{2m\pi}{3}, -\frac{1}{2}\right) ✓ \quad \text{and} \quad \left(\frac{\pi}{6} + \frac{(2m+1)\pi}{3}, \frac{1}{2}\right), ✓ \quad m \in \mathbb{Z}}\,.$$

[4]

(b) Determine the nature (i.e., point of local maximum/minimum, saddle point) of the critical points $\left(\frac{\pi}{3}, 0\right)$ and $\left(\frac{\pi}{6}, -\frac{1}{2}\right)$.

We will need

$$\boxed{f_{xx}(x,y) = -9y\sin(3x), \quad f_{xy}(x,y) = 3\cos(3x), \quad f_{yy}(x,y) = 2}\,. \quad 1/2\ ✓$$

We find that

$$D\left(\frac{\pi}{3}, 0\right) = \begin{vmatrix} f_{xx}\left(\frac{\pi}{3}, 0\right) & f_{xy}\left(\frac{\pi}{3}, 0\right) \\ f_{xy}\left(\frac{\pi}{3}, 0\right) & f_{yy}\left(\frac{\pi}{3}, 0\right) \end{vmatrix} = \begin{vmatrix} 0 & -3 \\ -3 & 2 \end{vmatrix} = -9 < 0\,. ✓$$

Thus, $(\pi/3, 0)$ is a saddle point. 1/2 ✓.

On the other hand,

$$D\left(\frac{\pi}{6}, -\frac{1}{2}\right) = \begin{vmatrix} f_{xx}\left(\frac{\pi}{6}, -\frac{1}{2}\right) & f_{xy}\left(\frac{\pi}{6}, -\frac{1}{2}\right) \\ f_{xy}\left(\frac{\pi}{6}, -\frac{1}{2}\right) & f_{yy}\left(\frac{\pi}{6}, -\frac{1}{2}\right) \end{vmatrix} = \begin{vmatrix} \frac{9}{2} & 0 \\ 0 & 2 \end{vmatrix} = 9 > 0\,. ✓$$

and $f_{xx}\left(\frac{\pi}{6}, -\frac{1}{2}\right) = \frac{9}{2} > 0$. 1/2 ✓ $\Big($or $f_{yy}\left(\frac{\pi}{6}, -\frac{1}{2}\right) = 2 > 0$. 1/2 ✓$\Big)$. Thus, $(\pi/6, -1/2)$ is a point of local minimum. 1/2 ✓

[4]

5. Let $D := \{(x,y) \in \mathbb{R}^2 \colon\ x^2 + y^2 \geq 1, \quad 0 \leq x \leq y, \quad y \leq 1\}$.

Set up the double integral $\displaystyle\iint_D \frac{xy^2}{(x^2+y^2)^2}\,dxdy$. in polar coordinates **without evaluating it**.

We use the polar coordinates. D is the domain lying in the first quadrant, outside the circle $x^2+y^2=1$, above the line $y=x$ and below the line $y=1$. Hence, the domain D is described in polar coordinates $r,\ \theta$ as

$$\left\{(r,\theta):\quad \frac{\pi}{4}\le\theta\le\frac{\pi}{2},\quad 1\le r\le\frac{1}{\sin\theta}\right\}.$$

Notice, $\frac{\pi}{4}\le\theta\le\frac{\pi}{2}$ is obtained from $0\le x\le y$, while $1\le r\le\frac{1}{\sin\theta}$ is obtained from $x^2+y^2\ge1$ and $y\le1$.
Therefore, recalling that the Jacobian when using polar coordinates is equal to r, we find that

$$\iint_D \frac{xy^2}{(x^2+y^2)^2}\,dxdy=\int_{\frac{\pi}{4}\ 1/2\ \checkmark}^{\frac{\pi}{2}\ 1/2\ \checkmark}\int_{1\ 1/2\ \checkmark}^{\frac{1}{\sin\theta}\checkmark}\frac{(r\cos\theta)(r\sin\theta)^2}{r^4}\,r\ \ 1/2\ \checkmark drd\theta\quad (\text{Notice } \checkmark \text{ for } \frac{1}{\sin\theta})$$

That full tick for $\frac{1}{\sin\theta}$ should be awarded only if there is some explanation. Otherwise, award only 1/2

[3]

Section B **Full Mark: 10**

1. The function $f(x,y):=-x^{2023}+y^{2021}$ has

   (a) a local maximum at $(0,0)$; (b) a local minimum at $(0,0)$;

   (c) a global minimum at $(0,0)$; (d) a saddle point at $(0,0)$;

   (e) a global maximum at $(0,0)$.

   Solution: (d)

   [2]

2. The double integral $\iint_T \sin(x+y)dxdy$ over $T:=\{(x,y)\in\mathbb{R}^2\colon\ 0\le x\le\pi/2,\ x\le y\le\pi/2\}$ is equal to

   (a) $\frac{1}{2}$; (b) 1; (c) -2; (d) -1; (e) 0.

   Solution: (b)

   [2]

3. The maximum increase of the function $f(x,y):=e^{x^2-y}\sqrt{y}$ at the point $(1,4)$ is equal to:

   (a) $e^{-3}\sqrt{16+\frac{49}{16}}$; (b) $16e^6$; (c) $\frac{3}{2}e^{-3}$; (d) $\frac{7}{4}e^{-3}$; (e) $\sqrt{e^{-3}\left(16+\frac{49}{16}\right)}$.

   Solution: (a)

   [2]

4. Find the cylindrical coordinates (r, θ, z) of the point $(\frac{\sqrt{3}}{4}, \frac{1}{4}, \frac{\sqrt{3}}{5})$.

(a) $(\frac{1}{4}, \frac{\pi}{6}, \frac{\sqrt{3}}{5})$; (b) $(\frac{1}{4}, \frac{2\pi}{3}, \frac{\sqrt{3}}{5})$; (c) $(\frac{1}{2}, \frac{\pi}{6}, \frac{1}{2})$; (d) $(\frac{1}{2}, \frac{\pi}{3}, \frac{\sqrt{3}}{5})$; (e) $(\frac{1}{2}, \frac{\pi}{6}, \frac{\sqrt{3}}{5})$.

Solution: (e)

[2]

5. Let $D := \{(x, y) \in \mathbb{R}^2 \colon x^2 + y^2 \leq 1, \quad x \leq 0, \quad x\sqrt{3} \leq y \leq -x\sqrt{3}\}$. If the description in polar coordinates of the domain D is given by $\{(r, \theta)\colon 0 \leq r \leq 1, \quad \theta_1 \leq \theta \leq \theta_2\}$ then

(a) $\begin{cases} \theta_1 = -\frac{\pi}{3}, \\ \theta_2 = \frac{\pi}{3}; \end{cases}$ (b) $\begin{cases} \theta_1 = -\frac{\pi}{3}, \\ \theta_2 = \frac{2\pi}{3}; \end{cases}$ (c) $\begin{cases} \theta_1 = \frac{2\pi}{3}, \\ \theta_2 = \frac{4\pi}{3}; \end{cases}$ (d) $\begin{cases} \theta_1 = \frac{2\pi}{3}, \\ \theta_2 = \frac{5\pi}{6}; \end{cases}$ (e) $\begin{cases} \theta_1 = \frac{2\pi}{3}, \\ \theta_2 = \frac{7\pi}{6}. \end{cases}$

Solution: (c)

[2]